- Complex policy sets can be effectively managed with mathematical precision.

In just a few months since its launch, Cedar Analysis has experienced rapid adoption among major technology firms. MongoDB, StrongDM, and others are already utilizing it in production to manage fine-grained authorization policies with confidence. This is more than just anecdotal; the symbolic compiler that supports Cedar Analysis is paired with formal proofs written in Lean, providing strong mathematical guarantees that the analysis results accurately reflect policy behavior at runtime. In practice, this means you can refactor policies, create new rules, or consolidate legacy permissions—and still be confident that your authorization logic remains sound and secure.

With SMT-based verification, your access control policies are more than code; they become formally validated artifacts. No more guesswork. No silent mistakes. Just proven correctness, ensuring your policies perform exactly as intended, today and in the future.

ANALYZABILITY

Because Cedar policies utilize a limited set of types and pure expressions, they can be analyzed statically with formal guarantees.

- Symbolic encoding of policies allows SMT-based checks for invariants, equivalence, or coverage. See the earlier sidebar “SMT analysis and why it matters for authorization” for more details on how this works.
- Cedar Analysis helps teams demonstrate that refactored policies maintain behavior.
- Policies can be checked for contradictions or unreachable rules before they are deployed in production.

Together, these capabilities ensure teams can confirm that policy changes do not create vulnerabilities before deployment.